

2212 Java Software Dev I

Project 2 – CFA Pay

Purpose

Practice input, variable declarations, assignment statements, calculations, String and Math class methods, and formatted output.

Goal

Write a program that breaks down an employee's pay into different denominations and creates formatted output.

CFA Currency Overview

The official currency of Central Africa is the CFA. Here's the description of CFA from Wikipedia.

"The Central African CFA franc (French: franc CFA or simply franc, ISO 4217 code: XAF) is the currency of six independent states in Central Africa: Cameroon, Central African Republic, Chad, Republic of the Congo, Equatorial Guinea and Gabon. These six countries have a combined population of 55.2 million people (as of 2020), and a combined GDP of US\$113.322 billion (as of 2020).

CFA stands for Coopération financière en Afrique centrale ("Financial Cooperation in Central Africa"). It is issued by the BEAC (Banque des États de l'Afrique Centrale, "Bank of the Central African States"), located in Yaoundé, Cameroon, for the members of the CEMAC (Communauté Économique et Monétaire de l'Afrique Centrale, "Economic and Monetary Community of Central Africa"). The franc is nominally subdivided into 100 centimes, but no centime denominations have been issued." - Wikipedia

Project Setting

Imagine you're putting your coding skills to work for a non-profit organization in Central Africa. The organization, based in Yaoundé—the capital of Cameroon—has about 50 employees.

Every two weeks, the accounting office handles payroll, which is distributed entirely in cash. This means someone must calculate exactly how many of each bill and coin are needed, then make a bank withdrawal for that amount. It's a tedious job for a human, but an ideal task for a computer.

Recognizing the need, you volunteer to develop a program that generates a report showing how many of each denomination are required to pay the employees.

To begin designing the solution, you gather the following requirements from the accounting department:

- The program must separately input the employee's first and last names.
- Pay will always be less than 100,000 CFA.
- The smallest denomination used is the 5 CFA coin.

- Calculations must start with the largest denomination (10,000 CFA) and work down to 5 CFA.

Before building the full system, you decide to create a prototype. This proof-of-concept program should prompt for an employee's name and salary (storing each in a variable), then generate a report like the following:

Name	Pay	10k	5k	1k	500	100	50	25	10	5
=====	=====	===	===	===	===	===	===	===	===	===
Metuge, Roggy	29890	2	1	4	1	3	1	1	1	1

Program Requirements

Your program should prompt the user to enter three pieces of information: an employee's first name, last name, and pay in CFA. After collecting this data, the program should generate output in the format described previously.

The employee's full name must be formatted as "LastName, FirstName" and displayed in a left-justified field that is 15 characters wide. Use `String` and `Math` class methods to either truncate the name to the first 15 characters or show the full name if it is shorter than 15 characters. This part of the solution must be implemented using only the knowledge covered thus far in the textbook—meaning no `if` statements or Boolean logic. *AI Tip:* Attempt a solution to this part of the problem first. Then do some research to see if there is a better way.

For clarity and organization, divide your program into three sections:

1. **Input:** Collect the employee's name and pay.
2. **Calculations:** Perform denomination breakdowns as needed and store values in variables.
3. **Output:** Display the formatted name and calculated results.

Here's a hint. Remember that an `int` divided by an `int` always gives us an `int` result. Once we calculate how many 10k bills we can get from the total pay, we need to calculate what's left before we calculate how many 5k bills we need. **What new Java operator have you learned that is perfect for calculating what's left over after we divide a number by another number?**

Expectations

The program is expected to compile successfully, include identification comments at the top, and produce output that exemplifies good grammar and spelling. Failure to meet these standards may result in point deductions.